

i) Give two industrial applications of HYDRATION.

Answer:

HYDRATION:- A chemical reaction in which a substance chemically combines with water is called hydration reaction. In organic chemistry, the addition of water to unsaturated substances (alkene or alkyne) is called hydration reaction. Addition of water to a compound or substance is commonly used in various industrial applications to achieve specific objectives. Here are two significant industrial applications of hydration:

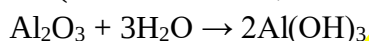
Cement Production: Hydration plays a fundamental role in the setting and hardening of cement. When water is added to cement powder, a series of exothermic chemical reactions occur, leading to the hydration of cement compounds such as tricalcium silicate (C3S), dicalcium silicate (C2S), tricalcium aluminate (C3A), and tetracalcium aluminoferrite (C4AF). The hydration process forms calcium silicate hydrates (C-S-H) and calcium hydroxide (CH) gels, which contribute to the binding and hardening of the cement paste.

Food and Beverage Industry:

Baking: Hydration is a key step in the formation of dough from flour in baking. When water is added to flour, proteins in the flour (such as gluten) hydrate and form a network that gives dough its elasticity, strength and texture of bread and other baked goods.

Production of Hydrated Lime: One significant application of hydration is in the production of hydrated lime (calcium hydroxide, Ca(OH)_2) from quicklime (calcium oxide, CaO). This process is commonly used in agriculture, and water treatment. $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

Production of Hydrated Alumina: The hydration is utilized in the production of hydrated alumina (aluminum hydroxide, Al(OH)_3) from alumina (aluminum oxide, Al_2O_3) in the aluminum industry.

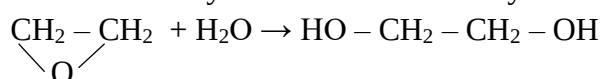


Hydrated alumina is processed to obtain aluminum metal by Bayer process and Hall-Héroult process.

Hydration in organic chemistry:

Ethylene Glycol:

Reaction of ethylene oxide with water yields ethylene glycol.



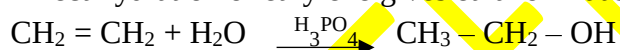
A large excess of water is supplied to the reactor to favour ethylene glycol formation over that of diethylene glycol and triethylene glycol, as the glycols are more reactive with ethylene oxide than is water.

Ethylene glycol is used as:

- i) antifreeze (Almost 43 percent of glycol is consumed in this category)
- ii) polyester fibre formation (Almost 26 percent of glycol is consumed in this process)

Ethanol:

Direct hydration of ethylene gives ethanol industrially.



Ethylene and high temperature steam are mixed and passed over an acidic catalyst, usually phosphoric acid on a support at 250 to 300°C under pressure.

Synthetic ethanol is used:

- i) as a chemical intermediate (for ethyl acetate, ethyl acrylate, glycol ethers, ethyl amines, etc.)
- ii) in toiletries and cosmetics
- iii) as a solvent
- iv) as a raw material for vinegar
- v) in household cleaners, in detergents, in pharmaceuticals, in printing inks etc.

ACETALDEHYDE:

Acetaldehyde was originally made by the hydration of acetylene over an oxidation–reduction catalyst, mercurous–mercuric sulphate buffered by ferric sulphate. Vinyl alcohol is assumed to form momentarily and rearrange to acetaldehyde at atmospheric pressure and 95°C.

